

## Title

Optimizing Implementation Strategies for a Comprehensive School-Based Obesity Prevention Program: A Cluster Randomized Hybrid Effectiveness-Implementation Trial Protocol

## Abstract

Evidence-based school interventions for obesity prevention often achieve incomplete implementation in real-world settings due to organizational and resource barriers. This protocol describes a hybrid type III cluster-randomized trial comparing two distinct implementation strategy bundles for delivering a comprehensive school-based healthy lifestyle program across multiple schools. Approximately 45 schools will be assigned to receive either a basic implementation support bundle or an enhanced bundle incorporating multiple implementation strategies including educational outreach, stakeholder engagement, and clinical-school linkages. Outcomes assess the degree to which intervention components are delivered with fidelity, measure parental responsiveness to health messaging, and track organizational factors influencing successful adoption. We seek to identify the most efficient and effective combination of implementation strategies to promote sustained program delivery in school settings.